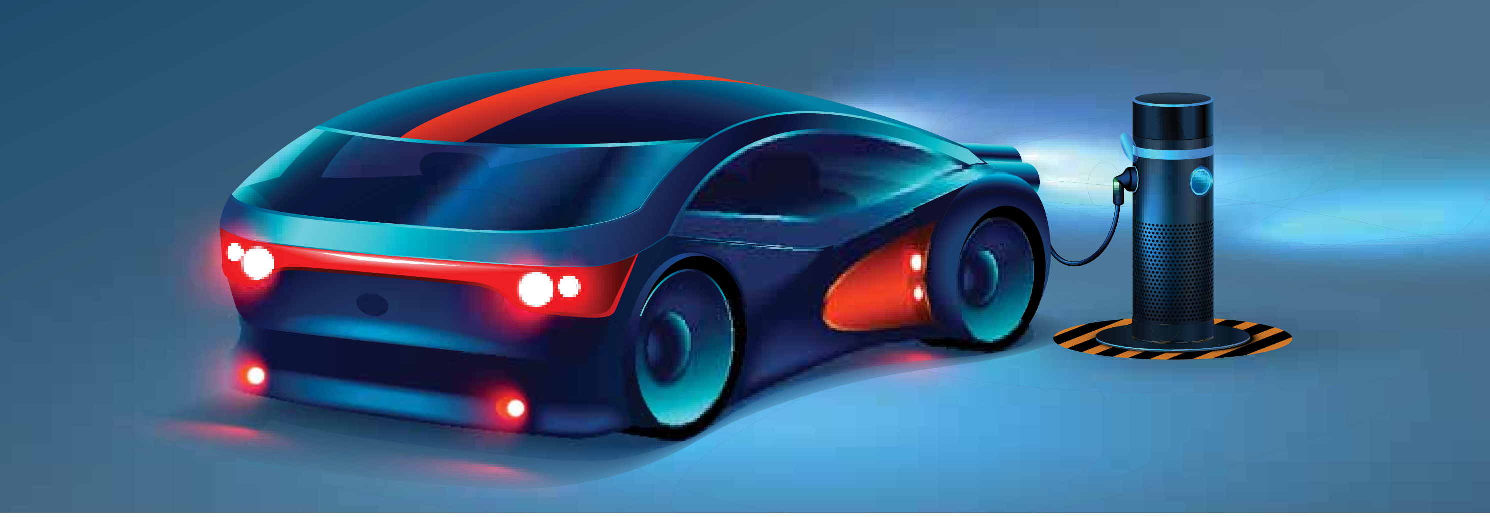
Challenges And Solutions For Next-Gen Automotives



Recent advancements in state-of-the-art technologies have increased the popularity of intelligent self-driving cars and electric vehicles. Yet their ability to face several demanding factors while on the road is fraught with complications that need to be resolved

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India is famous (or infamous) for having poor road conditions and traffic that can serve as a perfect testing ground for intelligent vehicles anywhere in the world. But the country is quite behind in their acceptance despite possessing experts with outstanding technical knowledge. At the same time, there has been significant move and investments towards smart precincts and smart cities over the recent past.

Meanwhile, electric vehicles (EVs) are gradually witnessing some degree of recognition among the general public thanks to extensive awareness initiatives by the government and large-scale support by various corporations.

Current EV trends

IC engines used to follow different categories of standards. Since shifting to electric vehicles, these standards have changed. Things have now moved on to higher voltage ranges of 72V to 700V from lower voltages like 12V and 24V. With this the required test infrastructure and facilities have entered a new phase, ushering in the era of EV adoption.

"With many more EVs to come on the road, the amount of feedback coming out from the vehicles in real-world scenarios will increase. This data is crucial for improving future system designs and increase the EV adoption," says Sujit Yardi, chief engineer, Devise Electronics.

AC charging stations in households can easily employ a 15A in-house plug for charging an EV. Several end-point delivery companies such as Amazon, Flipkart, and Swiggy use this model. So rather than a CAPEX model, people are going for an outsourced (or OPEX) model where the vehicle is not purchased but rented. In the coming years, the OPEX model is going to be more popular than the CAPEX model.

At present, consolidation of the standard protocols is taking place on a global scale for the EVs and charging stations so that a single charger can be used anywhere, preventing any confusion caused by the use of different charging standards. As of now Japan uses CHAdeMO while Tesla uses its proprietary connectors. We see CCS2 (combined charging system) being used in Nexon EV and MG ZS EV. Down the line, most of the manufacturers will migrate to CCS2 standard, which is compatible with both AC and DC charging stations/piles.

On the standards that EV makers have to comply with in India, the Automotive Research Association of India (ARAI) has formulated different